

PROFESSIONAL NURSING COACH

NURSING RESEARCH & STATISTICS

PNC ULTRA-LMR HYBRID MASTER COMPENDIUM • NUR-19 • VERIFIED

Process + designs + sampling + analysis + stats + publish • Batches 1–6 • 63 LMR pages + cover & TOC • NORCET / RRB / ESIC / DSSSB

Batch	Units	Pp
B1	I + II-A: research, EBP, problem, hypothesis, variables, lit	12
B2	III-B + IV-A: review, plagiarism, designs, bias	12
B3	V-A: sampling, tools, validity, ethics	12
B4	VI + VII-A: coding, tabulation, analysis, descriptives	12
B5	VII-B: dispersion, normal, tests, correlation, packages	10
B6	VIII: communicate, publish, utilize, ethics, predatory	5

WhatsApp DM: +91 90571 22020 • Hybrid: navy hero + chips + teal bars + Trap/Hit + Pearl + Speed-check • v1.0 Sep-2026

Pp	Block	Focus
4-15	I-II (B1)	Research, EBP, problem, hypothesis, variables, literature
16-27	III-IVa (B2)	Review, plagiarism, designs, bias, control, blinding
28-39	V-VIa (B3)	Sampling, tools, validity, reliability, ethics
40-51	VIIb-VIIa (B4)	Coding, tabulation, cleaning, analysis choice, descriptives
52-61	VIIb (B5)	Dispersion, normal, tests, errors, correlation, packages
62-66	VIII (B6)	Report, article, utilization, ethics, predatory, data

How to use: hero first, chips glance, table study, trap vs hit, pearl anchor, speed-check close. Double-QA pending sign-off per current guidance.

Q	Standard	Status
Q1	Units I-VIII mapped, batches 1-6	Done
Q2	Design/stats/ethics framings correct	Needs sign-off
Q3	One concept/page, fonts 10.5-21pt, 30% air	Done

Systematic + objective + empirical + verifiable + ethical + useful! Basic + applied + action + historical + survey + experimental + qualitative + nursing + priority + funded + published + used!

Systematic + empirical + ethical	Basic/applied/ action + nursing	Used + published + audit!
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★ WORD — MEANS

Word	Means
Research/nursing	Systematic + care + outcomes + EBP + audit
Basic/applied/action	Know + solve + change + ward + audit
Priority/funded	Gap + money + ethics + registry + date
Used/published	Journal + policy + bedside + audit + pride
⚠ FATAL TRAP 'Research theory' + no-question/method + 'PhD only' + bedside-zero + no-ethics/audit = shelf + misconduct + waste! Ask + method + ethics + use + audit.	★ NORCET HIT HIT: Nightingale + Cochrane + EBP + QI + research + audit + registry!

★ PNC PEARL: Asked systematically, answered honestly, used bedside.

► **Bedside-doubt + no-study + 'PhD only' + junior + curious. First trio?**

Question + method + ethics + mentor + audit.

NEED + AREAS OF NURSING RESEARCH

Gaps + outcomes + costs + safety + staffing + education + policy + priority + funds + audit!
Practice + education + admin + informatics + community + palliative + geriatric + child + mental + OBG!

Gaps + outcomes +
costs + safety

Practice/education/
admin/community + audit

Priority + funds
+ audit!

▲ NEED / AREA — EXAMPLES

Need / area	Examples
Need/importance	Gaps + outcomes + costs + safety + policy
Practice/admin/edu	Staffing + teaching + informatics + audit
Priority/funds	Thrust + grants + ethics + registry + date
Nurse	Ask + team + publish + use + audit + pride
⚠ FATAL TRAP Ritual + 'research burden' + no-question/time + bedside-zero + no-funds/ethics + no-use = shelf + burnout + waste! Time + mentor + funds + use + audit.	★ NORCET HIT HIT: ICMR + DST + DBT + WHO + Sigma + INC + priority + agenda + audit!

★ PNC PEARL: Bedside questions deserve funded answers.

► Falls + no-study + ward + repeated + junior + worried. Act trio?

Question + team + mentor + ethics + audit.

SCIENTIFIC METHOD + GOOD RESEARCH

Problem + review + hypothesis + design + sample + tools + collect + analyze + report + use + ethics + audit! Empirical + systematic + controlled + verifiable + ethical + generalizable + useful + published!

**Problem→review→
hypothesis→design**

**Sample/tools/collect/
analyze/report/use**

**Ethical + audited
+ useful!**

◆ STEP / TRAIT — DOES

Step / trait	Does
Steps	Problem + review + design + sample + report
Traits	Empirical + controlled + verifiable + ethical
Guard	Bias + fraud + plagiarism + audit + registry
Nurse	Follow + assist + use + publish + audit

⚠ FATAL TRAP

Jump-to-collect + no-review/design + 'common sense' + no-ethics/audit + fake-data = garbage + fraud + jail + retraction! Steps + ethics + audit.

★ NORCET HIT

HIT: Bacon + Popper + Kuhn + falsify + replicate + peer + audit!

★ PNC PEARL: Steps keep science honest.

► Collect-first + no-design + bias + junior + eager. Fix trio?

Steps + design + ethics + mentor + audit.

Identify + review + framework + hypothesis + design + sample + tools + pilot + collect + analyze + report + use + ethics + audit! Proposal + permission + funds + team + timeline + registry + publish!

Identify→review→
design→collect→use

Proposal + funds +
team + timeline

Ethics + registry
+ publish + audit!

Σ PHASE — DELIVERS

Phase	Delivers
Plan	Problem + review + design + proposal + funds
Do	Sample + tools + pilot + collect + audit
Analyze/report	Stats + tables + report + publish + registry
Use	EBP + policy + bedside + teach + audit

⚠ FATAL TRAP

Skip-review + no-proposal/funds + 'collect first' + no-pilot/ethics + lost-data + no-report/use = waste + fraud + audit-fail! Map + funds + ethics + audit.

★ NORCET HIT

HIT: Gantt + budget + team + roles + milestones + DSMB + audit!

★ PNC PEARL: Map first, collect second, use always.

► **Collect-first + no-proposal + no-ethics + junior + eager. Fix trio?**

Map + proposal + ethics + mentor + halt.

Best-evidence + expertise + patient-values + context + decision + outcomes + audit! Ask + acquire + appraise + apply + assess + audit (6 As!) + PICO + hierarchy + pre-appraised + bedside + policy!

**Evidence + expertise
+ values + context**

**6 As + PICO +
hierarchy + audit**

**Bedside + policy
+ outcomes!**

★ A – DOES

A	Does
Ask/acquire	PICO + search + databases + librarian + audit
Appraise/apply	Hierarchy + validity + pilot + bedside
Assess/audit	Outcomes + review + sustain + publish
Nurse	Champion + journal-club + protocol + audit

⚠ FATAL TRAP

'We always did' + no-search/appraisal + expert-only + patient-ignored + no-outcomes/audit = ritual + harm + waste! Search + appraise + apply + audit.

★ NORCET HIT

HIT: Cochrane + JBI + GRADE + hierarchy + RCT + SR + guideline + protocol + audit!

★ PNC PEARL: Evidence + expertise + preference = decision.

► **Ritual + no-search + 'always did' + ward + harm + junior. Fix trio?**

PICO + search + appraise + apply + audit.

Time + skills + access + English + support + culture + money + audit! Research (new!) + EBP (use!) + QI (improve!) + audit + outcomes + spread + sustain — know which lane you drive!

Time/skills/access/ support + audit	Research-new + EBP-use + QI-fix	Lane + outcomes + spread!
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▲ LANE — RUNS AS

Lane	Runs as
Research	New + method + ethics + publish + registry
EBP	Use + protocol + bedside + audit + sustain
QI	Fix + PDSA + run-charts + spread + audit
Barriers	Time + skills + access + support + money

⚠ FATAL TRAP

'Research = QI' + no-ethics/review + 'project' + no-measure/spread + no-audit = confusion + fraud + waste! Lane + ethics + measure + audit.

★ NORCET HIT

HIT: SQUIRE + STROBE + CONSORT + PRISMA + COREQ + EQUATOR + audit!

★ PNC PEARL: Right lane, right rules, right results.

► QI + no-ethics + 'research' + publish + junior + confused. Clarify trio?

Lane + ethics + measure + mentor + audit.

PROBLEM AREA + STATEMENT + CRITERIA

Bedside + gap + literature + feasibility + interest + novelty + ethics + fundable + significant + researchable + clear + concise + complete! Broad-area → focused-statement — funnel + feasibility + favor + funders!

**Bedside + gap +
literature + feasible**

**Funnel + focus +
clear + fundable**

**Significant +
ethical + audit!**

◆ ITEM — RULES

Item	Rules
Area/problem	Bedside + gap + literature + feasible + audit
Statement	Clear + concise + complete + variables + date
Criteria	Significant + novel + ethical + fundable
Nurse	Ask + mentor + team + diary + audit

⚠ FATAL TRAP

Vague + 'study nurses' + no-gap/literature + unfeasible + no-mentor/ethics = reject + waste + fail! Funnel + feasibility + mentor + audit.

★ NORCET HIT

HIT: FINER + PICOT + feasibility + interest + novelty + ethics + relevance + audit!

★ PNC PEARL: Funnel wide, focus sharp, fund wisely.

► **Vague + no-gap + unfeasible + junior + eager + stuck. Fix trio?**

Funnel + feasibility + mentor + literature.

PICO + clear + focused + measurable + achievable + relevant + time-bound + primary + secondary + exploratory + audit! Question births design; objectives steer analysis!

PICO + SMART + primary + secondary	Question→design; objectives→analysis	Audit + date + pride!
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Σ ITEM – WRITES AS

Item	Writes as
Question	PICO + clear + focused + feasible + audit
Objectives	SMART + primary + secondary + verbs + date
Link	Question→design + objectives→analysis + audit
Guard	Vague + double + leading + jargon + audit
⚠ FATAL TRAP Double-barreled + leading + jargon + 'study everything' + no-link/design + no-audit = reject + confusion + fail! SMART + link + audit.	★ NORCET HIT HIT: Bloom + verbs + KSA + levels + alignment + blueprint + audit!

★ PNC PEARL: Question sharp, objectives SMART.

► **Double + leading + vague + junior + stuck + blank. Fix trio?**

PICO + SMART + link + mentor + rewrite.

HYPOTHESIS (NULL VS RESEARCH)

Educated-guess + testable + directional + null + research + simple + complex + associative + causal + one/two-tailed + alpha + power! H0 (no-difference!) vs H1 (difference!) + p + decide + report!

H0 = no-difference;
H1 = difference!

Directional +
tails + alpha + power

Test + p +
decide + report!

★ TYPE – SAYS

Type	Says
Null/directional	No-difference + one-way + tails + alpha
Research/simple/complex	Difference + variables + test + report
Associative/causal	Relate + cause + design + senior + audit
Test	p + power + decide + retain/reject + audit

⚠ FATAL TRAP

No-hypothesis + 'prove' + p-hack + HARKing + no-alpha/power + overclaim = fraud + retraction + shame! Pre-register + power + report + audit.

★ NORCET HIT

HIT: Fisher + Neyman + Bayes + p + CI + effect + power + G*Power + registry!

★ PNC PEARL: Hypothesize humbly, test honestly.

► Prove + p-hack + no-power + junior + eager + publish. Fix trio?

Pre-register + power + honest-report + mentor.

VARIABLES + OPERATIONAL DEFINITIONS

Independent + dependent + extraneous + control + confound + mediate + moderate + define + measure + level + reliability + validity! Conceptual → operational — define to measure, measure to test!

IV/DV + extraneous/
confound + control

Conceptual→operational
+ level + tool

Define + measure
+ test + audit!

▲ VARIABLE — ROLE

Variable	Role
Independent/dependent	Cause + effect + manipulate + measure
Extraneous/confound	Noise + bias + control + match + audit
Mediate/moderate	Path + condition + analyze + senior
Define	Conceptual + operational + level + tool

⚠ FATAL TRAP

Fuzzy + 'satisfaction' + no-define/level/tool + confound-ignored + no-control = noise + wrong + retraction! Define + level + tool + control + audit.

★ NORCET HIT

HIT: Nominal + ordinal + interval + ratio + dummy + composite + index + scale + audit!

★ PNC PEARL: Undefined variables measure mirages.

► Fuzzy + no-tool + confound + junior + stuck + vague. Fix trio?

Define + level + tool + control + mentor.

LITERATURE REVIEW: MEANING + SOURCES

Gap + map + justify + design + tools + benchmarks + citations + ethics + audit! Books + journals + theses + reports + grey + experts + databases + libraries + alerts + manager + audit — read + map + cite!

Gap + map +
justify + design

Books/journals/
grey/experts + audit

Read + cite +
manage + audit!

◆ SOURCE — FINDS VIA

Source	Finds via
Books/journals/theses	Library + catalog + shelves + loan + audit
Grey/experts/reports	Govt + NGO + thesis + contact + audit
Databases/alerts	PubMed + CINAHL + alerts + manager
All	Keywords + limits + save + cite + audit

⚠ FATAL TRAP

Google-only + 'review done' + no-databases/grey + copy-paste + no-cite/manage = thin + plagiarized + reject + shame! Search + manage + cite + audit.

★ NORCET HIT

HIT: MeSH + Boolean + trunc + filters + PRISMA + snowball + hand + registry!

★ PNC PEARL: Search wide, save everything, cite honestly.

► Google-only + copy-paste + no-cite + junior + deadline + panic. Fix trio?

Databases + manager + cite + mentor + time.

SEARCHING + REVIEW METHOD + MATRIX

PICO + keywords + Boolean + MeSH + filters + screen + appraise + extract + synthesize + matrix + themes + gaps + audit! Table + compare + contrast + critique + conclude + cite!

**PICO + Boolean +
MeSH + filters**

**Screen + appraise +
extract + synthesize**

**Matrix + themes +
gaps + audit!**

Σ STEP – DOES

Step	Does
Search/screen	PICO + Boolean + filters + PRISMA + audit
Appraise/extract	Validity + form + dual + resolve + audit
Synthesize/matrix	Table + themes + gaps + model + audit
Write/cite	Compare + critique + conclude + audit

⚠ FATAL TRAP

Cherry-pick + 'supports me' + no-screen/appraisal + single-reviewer + no-matrix/gaps = bias + thin + reject!
Dual + matrix + gaps + audit.

★ NORCET HIT

HIT: CASP + JBI + GRADE + MMAT + data + software + librarian + audit!

★ PNC PEARL: Matrix maps; gaps grant topics.

► Cherry-pick + single + no-matrix + junior + stuck + thin. Fix trio?

Dual + matrix + gaps + librarian + time.

Read + skim + appraise + extract + compare + critique + thematic + chronological + theoretical + write + cite + audit! Summarize + evaluate + organize + argue-gap + link-study — review to reveal hole, not to heap papers!

**Read + appraise +
extract + compare**

**Thematic/chrono/
theory + argue-gap**

**Write + cite +
audit!**

★ MOVE — DOES

Move	Does
Read/appraise/extract	Skim + CASP + form + dual + resolve
Organize/compare/critique	Themes + table + strengths + flaws
Argue-gap/link	Hole + rationale + objectives + audit
Write/cite	Flow + headings + references + audit

⚠ FATAL TRAP

List-summaries + 'many studies' +
no-critique/synthesis/gap + copy + no-cite =
annotated-list + plagiarism + reject! Critique +
synthesize + gap + cite.

★ NORCET HIT

HIT: Funnel + hourglass + story + signpost +
topic-sentences + transitions + audit!

★ PNC PEARL: Critique creates the gap your study fills.

► **List + no-critique/gap + junior + thin + stuck. Fix trio?**

Critique + synthesize + gap + mentor + rewrite.

Cite + summarize + evaluate + reflect + relevance + audit! Author + year + aim + method + findings + critique + use — each source judged + logged + linked to study!

Cite + summarize + evaluate + reflect	Relevance + use + link + audit	Logged + linked + audit!
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▲ LINE — WRITES

Line	Writes
Cite/summarize	Author + aim + method + findings + audit
Evaluate/reflect	Strengths + flaws + bias + use + audit
Relevance/link	Gap + objectives + design + audit + date
All	Manager + file + backup + audit + pride
⚠ FATAL TRAP Copy-abstract + 'annotated' + no-evaluate/link + lost-files + no-audit = padding + plagiarism + reject! Judge + link + file + audit.	★ NORCET HIT HIT: Zotero + Mendeley + EndNote + tags + PDFs + notes + share + audit!

★ PNC PEARL: Judge every paper, link every judgment.

► Abstracts + no-judgment + lost-files + junior + mess. Fix trio?

Evaluate + link + manager + mentor + file.

Quote + paraphrase + cite + acknowledge + originality + similarity + permission + audit! Copy + patch + self + ghost + gift + salami + fabricate + falsify = misconduct + retraction + blacklist + jail!

Quote/paraphrase/ cite + originality	Similarity + permission + audit	Misconduct = retract + blacklist!
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◆ ACT – RULES

Act	Rules
Quote/paraphrase/cite	Marks + own-words + style + audit + date
Similarity/permission	Check + threshold + rights + letter + audit
Misconduct-forms	Copy/patch/self/ghost/gift/salami + audit
Guard	Teach + software + policy + audit + pride

⚠ FATAL TRAP

Copy-paste + 'common knowledge' + no-quotes/cites + paid-thesis + ghost + no-check = expel + retract + blacklist + jail! Original + cite + check + audit.

★ NORCET HIT

HIT: Turnitin + iThenticate + UGC + COPE + ICMJE + Vancouver + APA + audit!

★ PNC PEARL: Cite what you didn't create.

► Copy-paste + no-cite + similarity-high + junior + panic. Fix trio?

Paraphrase + cite + check + rewrite + mentor.

QUANTITATIVE VS QUALITATIVE

Numbers + causes + generalize + stats + large + deductive + objective + audit! vs meanings + lived + depth + small + inductive + subjective + trust + audit — question picks paradigm, paradigm picks methods!

Numbers/causes/ generalize + stats	Meanings/lived/ depth + trust	Question picks paradigm + audit!
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Σ FACE — QUANT — QUAL

Face	Quant	Qual
Aim/sample	Cause + large + random	Mean + small + purposive
Data/analysis	Scales + stats + p + CI	Words + themes + quotes
Rigor	Validity + reliability + power	Trust + audit + reflexivity
Write	Tables + graphs + p + audit	Narrative + audit + pride

⚠ FATAL TRAP

'Quant superior' + qual-without-rigor + mixed-muddle + wrong-paradigm + no-audit = bias + thin + reject! Fit + rigor + audit.

★ NORCET HIT

HIT: Positivist + constructivist + critical + pragmatic + mixed + triangulation + audit!

★ PNC PEARL: Fit paradigm to question, rigor to paradigm.

► Meanings + small + depth + stories + nurse + curious. Paradigm + rigor?

Qual + trust + audit + themes + quotes.

Past + archives + documents + artifacts + oral + authenticity + criticism + context + narrative + lessons + audit! External + internal criticism + triangulation + bias + presentism + ethics + audit — past explains present practice!

Archives + oral +
artifacts + audit

External/internal +
criticism + context

Lessons + audit
+ pride!

★ STEP — DOES

Step	Does
Collect	Archives + oral + artifacts + audit + date
Criticize	External + internal + bias + context
Narrate/lessons	Story + themes + present + audit
Guard	Presentism + gaps + ethics + audit + pride

⚠ FATAL TRAP

Wikipedia + 'history' + no-archives/criticism + present-judgment + no-audit = myth + anachronism + reject! Archives + criticism + context + audit.

★ NORCET HIT

HIT: Nightingale + Seacole + Anandibai + Rath + oral + museums + diaries + audit!

★ PNC PEARL: Sources criticized, context crowned.

► Wikipedia + no-archives + present-judgment + junior + thin. Fix trio?

Archives + criticism + context + mentor.

Ask + sample + questionnaire + interview + phone + online + response + bias + pilot + ethics + analyze + report + use + audit! Descriptive + KAP + needs + satisfaction + census + poll + audit — ask well, generalize carefully!

Ask + sample +
tool + pilot + ethics

Response + bias +
analyze + report

Generalize + use
+ audit!

▲ STEP — DOES

Step	Does
Ask/sample/tool	Question + random + valid + pilot + ethics
Collect/response	Phone/online/face + chase + bias + audit
Analyze/report	Weight + tables + CI + report + use
Guard	Nonresponse + leading + fatigue + audit + date

⚠ FATAL TRAP

Convenience-only + 'survey done' + no-pilot/validity + low-response + no-weighting + overclaim = bias + junk + reject! Random + pilot + weight + audit.

★ NORCET HIT

HIT: NFHS + DLHS + NSS + census + exit + satisfaction + KAP + audit!

★ PNC PEARL: Response decides representativeness.

► **Convenience + low-response + overclaim + junior + quick. Fix trio?**

Random + pilot + weight + honesty + audit.

Cause + manipulate + control + randomize + blind + pre/post + CONSORT + ethics + registry + audit! RCT + gold + efficacy + NNT + harm + intention + per-protocol + fidelity + audit — randomize + blind + report!

**Manipulate + control
+ randomize + blind**

**CONSORT + ethics +
registry + audit**

**Gold + NNT +
harm + fidelity!**

◆ PILLAR — HOLDS BY

Pillar	Holds by
Manipulate/control/random	Cause + compare + chance + audit
Blind/pre/post	Mask + baseline + follow + fidelity
Ethics/registry	Consent + CTIR + DSMB + SAE + audit
Report/use	CONSORT + NNT + harm + guideline + audit

⚠ FATAL TRAP

No-consent/registry + 'trial' + unblinded + no-fidelity + harm-hidden + no-report = abuse + fraud + jail + retraction! Consent + registry + blind + audit.

★ NORCET HIT

HIT: Placebo + sham + dose + cluster + crossover + factorial + N-of-1 + pragmatic + audit!

★ PNC PEARL: Randomize, blind, register, report.

► **Trial + no-consent/registry + unblinded + junior + eager. Fix trio?**

Halt + consent + registry + blind + mentor.

Cause + no-random + control-group + pre/post + time-series + nonequivalent + threats + match + stats + ethics + audit! Real-world + feasible + weaker + careful — quasi needs extra caution + controls!

No-random +
control + pre/post

Threats + match +
stats + audit

Feasible + careful
+ honest!

Σ TYPE — USES

Type	Uses
Nonequivalent/control	Wards + match + pre/post + adjust
Time-series/interrupted	Many-points + trend + policy + audit
Threats/guards	History + maturation + testing + control
Report	STROBE/TREND + limits + honesty + audit
⚠ FATAL TRAP 'Experimental' + no-random + causal-claim + no-threats + no-audit = overclaim + reject + shame! Name + threats + adjust + honest + audit.	★ NORCET HIT HIT: Regression-discontinuity + DID + ITS + synthetic + propensity + audit!

★ PNC PEARL: Quasi claims quasi-causality, honestly stated.

► **No-random + causal-claim + no-threats + junior + bold. Fix trio?**

Name + threats + adjust + honesty + mentor.

One-shot + one-group + static-group + no-control/random + weak + pilot + hypothesis + feasibility + audit! Explore + test-tools + train + honest-limits + upgrade-next — start small, claim smaller!

One-shot/group +
no-control + weak

Pilot + feasibility
+ hypothesis + audit

Claim small +
upgrade next!

★ TYPE — USES

Type	Uses
One-shot/group	Explore + describe + hypothesis + audit
Static-group	Compare + no-random + caution + audit
Guard	Weak + confound + no-claim + honest + audit
Next	Pilot + tools + team + upgrade + date

⚠ FATAL TRAP

Causal-claim + 'study proves' + no-control + n-tiny + no-limits + publish-rush = junk + retraction + shame!
Limits + honesty + upgrade + audit.

★ NORCET HIT

HIT: Feasibility + acceptability + fidelity + dose + retention + stop-go + audit!

★ PNC PEARL: Pilot's job is learning, not proving.

► **One-group + proves + tiny + junior + rush + publish. Fix trio?**

Limits + honesty + upgrade + mentor + pilot.

DESCRIPTIVE + CORRELATIONAL DESIGNS

Describe + relate + predict + survey + case + records + r + regression + confound + longitudinal + audit! Association + strength + direction + significance — relate + predict, never prove cause!

Describe + relate
+ predict + survey

r + regression +
confound + audit

Associate, never
prove cause!

▲ TYPE — USES

Type	Uses
Descriptive/case	Who/what/where + records + audit
Correlational/predictive	r + regression + confound + audit
Longitudinal/cross	Time + snapshot + trend + audit
Guard	Cause-claim + third-variable + audit + honesty
⚠ FATAL TRAP Correlation + 'X causes Y' + no-confound/longitudinal + overclaim + no-audit = myth + retraction + shame! Associate + adjust + honest + audit.	★ NORCET HIT HIT: Pearson + Spearman + Kendall + point-biserial + phi + partial + multiple + audit!

★ PNC PEARL: Correlation courts, causation needs design.

► r-high + 'causes' + cross + junior + bold + press. Fix trio?

Associate + confound + design + honesty.

COHORT + PROSPECTIVE/RETROSPECTIVE

Exposure → outcome + forward + RR + incidence + time + costly + loss + audit! Retrospective (records + quick + bias!) + ambidirectional + nested + registry + ethics + audit — direction decides design + diligence!

Exposure → outcome +
forward + RR

Records + quick +
bias + registry

Direction + diligence
+ audit!

◆ TYPE — USES

Type	Uses
Prospective/cohort	Forward + RR + incidence + time + audit
Retrospective/records	Quick + bias + registry + audit + date
Nested/registry	Efficient + stored + link + audit + pride
Guard	Loss + cost + ethics + audit + honesty
⚠ FATAL TRAP Retrospective + 'cohort proves' + no-bias/loss + incomplete-records + overclaim = wrong + retraction + shame! Direction + bias + honesty + audit.	★ NORCET HIT HIT: Framingham + Nurses + Rothman + induction + latency + person-years + audit!

★ PNC PEARL: Forward follows risk; backward beware bias.

► Records + quick + 'proves' + bias + junior + bold. Fix trio?

Direction + bias + honesty + senior + audit.

BIAS, CONTROL, BLINDING, RANDOMIZATION

Selection + performance + detection + attrition + reporting + confounding + Hawthorne + placebo + allocation + conceal + block + stratify + audit! Control + blind + randomize + conceal + analyze-blind + report!

Selection/performance/ detection/attrition + audit	Blind + random + conceal + stratify	Control + report + audit!
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Σ THREAT – TAMED BY

Threat	Tamed by
Selection/attrition	Random + conceal + ITT + follow + audit
Performance/detection	Blind + placebo + objective + audit
Reporting/confound	Register + adjust + match + audit + date
All	Protocol + monitoring + DSMB + audit + pride

⚠ FATAL TRAP

No-random/blind + 'controlled' + dropout-ignored + outcome-switch + no-register + overclaim = bias + fraud + retraction! Register + blind + ITT + audit.

★ NORCET HIT

HIT: RoB + ROBINS + Jadad + allocation + sequence + envelopes + audit!

★ PNC PEARL: Blind the eyes that measure.

► No-blind + dropout-ignored + switch + junior + publish. Fix trio?

Blind + ITT + register + honesty + mentor.

Target + accessible + sample + inclusion + exclusion + attrition + consent + ethics + audit! Who + whom + how-many + how-chosen + how-protected — population promises, sample speaks, criteria guard!

Target/accessible/ sample + criteria	Who/whom/many/ chosen/protected	Inclusion + exclusion + audit!
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★ TERM — MEANS — ACT

Term	Means	Act
Target/accessible/sample	All + reach + chosen + consent + audit	
Inclusion/exclusion	In + out + screen + document + audit	
Attrition/dropout	Lost + reasons + ITT + follow + audit	
All	Ethics + registry + date + pride + audit	

⚠ FATAL TRAP

'Available + willing' + no-criteria/consent + dropout-ignored + 'sample enough' + no-audit = bias + coercion + junk + suit! Criteria + consent + follow + audit.

★ NORCET HIT

HIT: Universe + element + frame + unit + respondent + informant + gatekeeper + audit!

★ PNC PEARL: Criteria gatekeep generalizability.

► Available-only + no-criteria + dropout-ignored + junior + quick. Fix trio?

Criteria + consent + follow + audit + mentor.

Random + systematic + stratified + cluster + multistage + PPS + equal-chance + generalize + audit! Frame + start + interval + strata + clusters + stages + weight + ethics + audit — chance chooses, stats generalize!

Random/systematic/ stratified/cluster	Equal-chance + frame + weight	Generalize + audit!
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▲ TYPE — DOES

Type	Does
Simple/systematic	Random + interval + frame + audit + date
Stratified/cluster	Strata + groups + stages + weight + audit
Multistage/PPS	Many + size + cost + design-effect + audit
All	Equal + ethics + audit + pride + registry

⚠ FATAL TRAP

'Random' + haphazard + no-frame/weight + cluster-ignored + overclaim + no-audit = bias + wrong-CI + reject! Frame + method + weight + audit.

★ NORCET HIT

HIT: Lottery + tables + computer + K + strata + PSU + design-effect + audit!

★ PNC PEARL: Frame first, chance second, weight always.

► Haphazard + 'random' + no-frame + overclaim + junior. Fix trio?

Frame + method + weight + honesty + mentor.

Convenience + purposive + quota + snowball + consecutive + volunteer + expert + theoretical + audit! Cheap + quick + biased + honest-limits + pilot + qualitative + hard-to-reach + audit — convenience courts caution!

Convenience/purposive/ quota/snowball + audit	Cheap + quick + biased + honest	Pilot/qualitative/ hidden + audit!
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◆ TYPE — USES — LIMITS

Type	Uses	Limits
Convenience/quota/volunteer	Quick + cheap + pilot	Bias + no-generalize
Purposive/expert/theoretical	Informants + saturation + audit	
Snowball/consecutive	Hidden + clinic + chain + audit	
All	Honest-limits + audit + upgrade + date	

⚠ FATAL TRAP

Convenience + 'representative' + no-limits + generalize + no-audit = bias + wrong + reject + shame! Name + limits + honesty + audit.

★ NORCET HIT

HIT: Deviant + typical + critical + maximum-variation + homogeneous + confirm + audit!

★ PNC PEARL: Name it non-random, claim it non-general.

► Convenience + generalize + no-limits + junior + bold. Fix trio?

Name + limits + honesty + upgrade + mentor.

Sampling + nonsampling + coverage + nonresponse + measurement + processing + power + effect + alpha + attrition + audit! Size = design + variance + effect + power + alpha + dropout + budget + ethics!

Sampling/non-response/measure + audit	Power + effect + alpha + dropout	Size + budget + ethics + audit!
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Σ ITEM – RULES

Item	Rules
Errors/bias	Sampling + nonresponse + measure + audit
Size/power	Design + variance + effect + power + alpha
Adjust/dropout	Inflate + attrition + budget + ethics
Report	N + power + limits + audit + date + pride

⚠ FATAL TRAP

n=30-magic + 'enough' + no-power/effect + dropout-ignored + underpowered + overclaim = waste + false + reject! Calculate + justify + audit.

★ NORCET HIT

HIT: G*Power + PASS + nMaster + thumb + saturation + information-power + audit!

★ PNC PEARL: Powered samples speak; tiny ones whisper.

► n-magic + no-power + underpowered + junior + publish. Fix trio?

Calculate + justify + honesty + mentor.

Why + what + whom + when + where + how + how-long + how-much + who-collects + train + pilot + ethics + audit! Plan + tools + team + transport + money + calendar + backup + audit — plan collects quality!

Why/what/whom/ when/where/how + audit	Team + train + pilot + calendar	Backup + budget + audit!
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★ Q – ANSWERS

Q	Answers
Why/what/whom	Objectives + variables + sample + consent
When/where/how	Schedule + site + tools + team + audit
Pilot/train/backup	Test + skills + spare + money + audit
All	Plan + Gantt + budget + ethics + audit + date

⚠ FATAL TRAP

Collect-first + 'figure later' + no-plan/pilot/team + tools-broken + data-lost + no-backup = garbage + repeat + waste! Plan + pilot + backup + audit.

★ NORCET HIT

HIT: SOP + manual + training + certification + monitoring + DSMB + audit!

★ PNC PEARL: Planned collection is half analysis.

► **Collect-first + tools-broken + data-lost + junior + panic. Fix trio?**

Halt + plan + pilot + backup + mentor.

Clear + short + closed/open + scales + order + sensitive-last + pretest + reliability + validity + cover + consent + return + audit! Leading + double + jargon + long + threatening = lies + blanks + bias!

Clear + short +
scales + order

Sensitive-last +
pretest + consent

Return + audit
+ pride!

▲ PART — RULES

Part	Rules
Items/order	Clear + short + scales + sensitive-last
Pretest/pilot	Trial + revise + reliability + validity
Administer/return	Cover + consent + chase + audit + date
Guard	Leading/double/jargon + bias + audit + honesty
⚠ FATAL TRAP Long + leading + jargon + no-pretest/consent + 'fill fast' + no-chase + fake-fill = garbage + fraud + reject! Pretest + consent + chase + audit.	★ NORCET HIT HIT: Likert + VAS + semantic + Guttman + dichotomous + ranking + open + audit!

★ PNC PEARL: Pretested questions get truthful answers.

► **Leading + long + no-pretest + fake-fill + junior + rush. Fix trio?**

Revise + pretest + consent + chase + audit.

Rapport + private + probe + listen + record + time + thanks + follow-up + safety + audit!
Structured + semi + unstructured + focus-group + key-informant + phone + guide + consent + language + audit!

**Rapport + private
+ probe + listen**

**Structured/semi/
FGD/KII + guide**

**Record + thanks
+ audit!**

◆ TYPE – USES

Type	Uses
Structured/semi	Guide + probe + record + audit + date
Unstructured/FGD/KII	Explore + group + expert + themes
Phone/online	Reach + record + consent + audit + pride
Guard	Bias + leading + fatigue + safety + audit
⚠ FATAL TRAP Interrogate + lead + record-without-consent + unsafe-alone + no-probe/guide + lost-notes = lies + harm + suit + junk! Rapport + guide + consent + audit.	★ NORCET HIT HIT: In-depth + narrative + life-history + elite + cognitive + debrief + member-check + audit!

★ PNC PEARL: Probe gently, record faithfully.

► Lead + no-guide + unsafe-alone + lost-notes + junior. Fix trio?

Guide + consent + safety + record + mentor.

OBSERVATION + RECORDS/PHYSIOLOGICAL

Participant + non + structured + checklist + time + blind + reactivity + ethics + audit! Charts + registers + labs + vitals + devices + caliber + zero + repeat + audit — watch + measure + record + verify!

**Participant/non +
checklist + blind**

**Charts/registers/
labs/devices + audit**

**Caliber + repeat
+ verify!**

Σ TOOL — RULES

Tool	Rules
Observation	Checklist + time + blind + ethics + audit
Records/documents	Charts + registers + extract + verify
Physiological	Caliber + zero + repeat + audit + date
Guard	Reactivity + Hawthorne + drift + audit + honesty

⚠ FATAL TRAP

Watch-without-consent + 'objective' +
no-checklist/reliability + device-uncalibrated +
copy-records + no-verify = bias + breach + junk!
Consent + calibrate + verify + audit.

★ NORCET HIT

HIT: Event + time + ABC + field-notes + video +
inter-rater + audit!

★ PNC PEARL: Calibrated eyes and instruments both.

► Watch + no-consent + uncalibrated + copy + junior. Fix trio?

Consent + checklist + calibrate + verify.

Tick + rate + anchor + pilot + reliability + validity + train + audit! GCS + Braden + Morse + pain + Apgar + Partograph + MEWS + NEWS + audit — tick + rate + total + act + handover!

Tick + rate + anchor + pilot	GCS/Braden/Morse/ pain/MEWS + audit	Total + act + handover!
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★ TOOL — USES

Tool	Uses
Checklist	Tick + present/absent + audit + date + pride
Rating/likert/VAS	Anchor + pilot + reliability + audit
Clinical	GCS + Braden + Morse + MEWS + action
Guard	Halo + central + leniency + train + audit
⚠ FATAL TRAP Tick-without-observe + 'all normal' + no-anchor/pilot + missed deterioration + no-action + no-audit = fraud + death + suit! Observe + rate + act + audit.	★ NORCET HIT HIT: Cronbach + kappa + ICC + CVI + pilot + train + audit!

★ PNC PEARL: Ticked truthfully triggers timely action.

► All-normal + no-observe + missed-fall + junior + routine. Fix trio?

Observe + rate + act + senior + audit.

Face + content + criterion + construct + convergent + discriminant + known-groups + factor + CVI + experts + pilot + audit! Measures-what-claims + truth + use — valid tool, true results!

Face/content/ criterion/construct	CVI + experts + pilot + audit	True + useful + audit!
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▲ TYPE — PROVES BY

Type	Proves by
Face/content	Experts + CVI + relevance + audit + date
Criterion	Gold + concurrent + predictive + audit
Construct	Convergent + discriminant + factor + audit
All	Pilot + revise + report + audit + pride
⚠ FATAL TRAP 'Valid' + no-experts/CVI/pilot + borrowed-foreign + mistranslate + no-report = invalid + wrong + reject! Validate + report + audit.	★ NORCET HIT HIT: Lawshe + Lynn + kappa + ROC + sensitivity + AUC + audit!

★ PNC PEARL: Validity proven, not proclaimed.

► **Borrowed + no-CVI/pilot + mistranslate + junior + rush. Fix trio?**

Experts + CVI + pilot + report + mentor.

RELIABILITY + VALIDITY VS RELIABILITY

Stable + consistent + repeatable + Cronbach + test-retest + split + parallel + KR + kappa + ICC + SEM + audit! Reliable-necessary-not-sufficient + valid-sufficient + target + audit — consistent + true together!

**Stable + Cronbach +
retest + kappa + audit**

**Reliable→necessary;
valid→sufficient**

**Target + audit
+ pride!**

◆ TYPE — TESTS BY

Type	Tests by
Stability/equivalence	Test-retest + parallel + split + audit
Internal/homogeneity	Cronbach + KR + item + audit + date
Rater/objective	Kappa + ICC + blind + train + audit
Vs-validity	Reliable-target + valid-bullseye + audit
⚠ FATAL TRAP Alpha-only + 'reliable' + no-retest/rater + drift + no-audit + overclaim = shaky + wrong + reject! Test + report + audit.	★ NORCET HIT HIT: Target-analogy + dartboard + SEM + CI + floor/ceiling + audit!

★ PNC PEARL: Consistent arrow, true bullseye — need both.

► Alpha-only + no-retest + drift + junior + claim. Fix trio?

Retest + rater + report + honesty + mentor.

Consent + assent + risk/benefit + privacy + confidentiality + justice + vulnerable + EC + CTIR + compensation + audit! ICMR + Helsinki + Belmont + Nuremberg + GCP + DCGI + insurance + SAE + audit!

**Consent + risk +
privacy + justice**

**EC + CTIR +
compensation + audit**

**Vulnerable +
insurance + SAE!**

Σ PILLAR – DOES

Pillar	Does
Consent/assent	Inform + voluntary + child + witness + audit
Risk/justice/privacy	Minimize + fair + lock + audit + date
EC/CTIR/compensation	Review + register + pay + SAE + audit
Vulnerable	Extra + guardian + advocate + audit + pride
⚠ FATAL TRAP No-consent/EC + 'audit only' + coercion + breach + no-compensation/SAE + lost-records = abuse + jail + blacklist + shame! Consent + EC + audit.	★ NORCET HIT HIT: Tuskegee + Nazi + Beecher + Willowbrook + Milgram + Zimbardo + audit!

★ PNC PEARL: Consent + committee + compensation = ethical triad.

► No-EC + coercion + breach + junior + eager + vulnerable. Act trio?

Halt + EC + consent + advocate + audit.

Rehearse + test-tools + estimate + train + troubleshoot + refine + report + go/no-go + audit! Plan + team + transport + money + calendar + backup + field-control + supervision + audit — pilot predicts problems, procedure prevents them!

Rehearse + test +
estimate + refine

Team + transport +
money + backup

Go/no-go +
audit + date!

★ ITEM — DOES

Item	Does
Pilot	Tools + time + cost + issues + refine + audit
Team/field	Train + transport + money + backup + audit
Control/supervise	Spot + back-check + diary + review
All	Report + go/no-go + audit + pride + date

⚠ FATAL TRAP

Skip-pilot + 'main will manage' + tools-broken + cost-blast + no-backup/supervision = garbage + repeat + waste! Pilot + backup + supervise + audit.

★ NORCET HIT

HIT: 10% + stop-go + fidelity + acceptability + retention + audit!

★ PNC PEARL: Pilot small, main confident.

► No-pilot + tools-broken + cost-blast + junior + panic. Fix trio?

Pilot + backup + supervise + mentor + halt.

DATA PREPARATION + CODING + CLASSIFICATION

Edit + code + classify + enter + verify + backup + audit! Codebook + numbers + missing + wild + double-entry + range-checks + lock + version + audit — clean data in, credible results out!

**Edit + code +
enter + verify**

**Codebook + missing
+ range + lock**

**Backup + version
+ audit!**

▲ STEP — DOES

Step	Does
Edit/code/enter	Clean + numbers + double + range + audit
Classify/verify	Groups + back-check + wild + audit + date
Backup/version	Copy + cloud + log + lock + audit + pride
Guard	Fabricate + loss + breach + audit + action

⚠ FATAL TRAP

Fabricate + 'adjust outliers' +
no-codebook/verify/backup + lost-pen-drive + breach =
fraud + loss + jail + retraction! Codebook + verify +
backup + audit.

★ NORCET HIT

HIT: EDC + REDCap + ODK + KoBo + range + skip +
audit-trail + audit!

★ PNC PEARL: Codebook today, credibility tomorrow.

► **No-codebook + lost-drive + no-backup + junior + tears. Fix trio?**

Codebook + backup + verify + audit + mentor.

Rows + columns + stubs + captions + footnotes + totals + source + audit! Tallies + tables + master + cross + pivot + charts + draft + review + audit — tables talk before text speaks!

Rows/columns/stubs/ captions/footnotes	Master + cross + pivot + charts	Source + audit + pride!
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◆ TABLE — RULES

Table	Rules
Structure	Rows + columns + stubs + captions + audit
Master/cross/pivot	Tallies + cross + pivot + charts + audit
Source/footnotes	Units + totals + source + date + audit
Guard	Fudge + cherry + hide + audit + honesty
⚠ FATAL TRAP Fudged + 'presentable' + no-source/footnotes + cherry + hidden-missing + no-audit = fraud + retraction + shame! Source + honesty + audit.	★ NORCET HIT HIT: APA + AMA + CONSORT + STROBE + PRISMA + tables + figures + legends + audit!

★ PNC PEARL: Footnoted tables testify truthfully.

► **Fudged + no-source + cherry + junior + pressured. Fix trio?**

Source + honesty + audit + mentor + redo.

Missing + outliers + wild + logic + range + impute + sensitivity + document + audit! Meaning + compare + explain + limits + implications + recommend + audit — clean first, claim later, confess limits always!

Missing/outliers/ wild + logic + audit	Impute + sensitivity + document + audit	Interpret + limits + recommend!
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Σ STEP – DOES

Step	Does
Clean/missing	Wild + logic + range + impute + audit
Check/sensitivity	Double + outlier + robust + document
Interpret/compare	Meaning + literature + limits + audit
Recommend	Implications + practice + research + audit

⚠ FATAL TRAP

Delete-missing + 'outliers removed' + no-document/sensitivity + over-interpret + no-limits = bias + fraud + retraction! Document + sensitivity + honesty + audit.

★ NORCET HIT

HIT: MAR/MCAR/MNAR + multiple + LOCF + per-protocol + ITT + audit!

★ PNC PEARL: Documented cleaning beats hidden polishing.

► **Delete-missing + no-document + overclaim + junior + rush. Fix trio?**

Document + sensitivity + honesty + mentor.

Objectives + design + variables + levels + distribution + sample + assumptions + tests + software + statistician + audit! Right-test + right-graph + right-report + right-claim — analysis answers objectives, nothing more!

**Objectives + design
+ levels + distribution**

**Right-test + graph
+ report + claim**

**Statistician +
audit + date!**

★ Q – POINTS TO

Q	Points to
Objectives/design/levels	t/chi/ANOVA + ranks + regression + audit
Distribution/sample	Normal + n + transform + nonparam + audit
Software/statistician	SPSS/R + consult + code + audit + date
Guard	Fishing + p-hack + HARKing + audit + honesty

⚠ FATAL TRAP

t-test-everything + 'p-significant' + fishing + no-assumptions/statistician + overclaim = wrong + retraction + shame! Consult + check + honest + audit.

★ NORCET HIT

HIT: Flowchart + decision-tree + handbook + consult + review + audit!

★ PNC PEARL: Objectives choose tests, not habits.

► **t-everything + fishing + no-assumptions + junior + publish. Fix trio?**

Consult + check + honesty + decision-tree.

Collect + organize + summarize + analyze + interpret + present + decide + predict + audit! Counts + rates + trends + trials + quality + policy + bedside + management + research + audit — data + decisions + delight!

**Collect→present +
decide + predict**

**Counts/rates/trends/
trials/quality + audit**

**Bedside + policy
+ pride!**

▲ USE — NURSING DOES

Use	Does
Describe/decide	Summarize + compare + choose + audit
Predict/policy	Trend + trial + forecast + budget + audit
Quality/research	Run-charts + trials + publish + audit
Nurse	Collect + calculate + interpret + teach + audit

⚠ FATAL TRAP

'Stats scary' + avoid + no-collect/calculate + gut-decisions + no-audit = error + waste + harm! Learn + calculate + interpret + audit.

★ NORCET HIT

HIT: Nightingale + Farr + Graunt + Petty + Quetelet + Galton + Pearson + Fisher + audit!

★ PNC PEARL: Numbers narrate nursing needs.

► Gut-only + no-numbers + error + ward + audit-query. Fix trio?

Collect + calculate + interpret + teach.

Describe (mean/median/mode + SD + graphs!) + infer (tests + CI + p + generalize!) + estimate + hypothesize + power + sample + assumptions + audit! Sample describes, inference leaps — with CI + caution!

**Describe + graphs
+ mean/SD + audit**

**Infer + tests +
CI/p + generalize**

**Leap + caution
+ audit!**

◆ BRANCH — DOES

Branch	Does
Descriptive	Mean/median/mode + SD + graphs + audit
Inferential	Tests + CI + p + generalize + audit
Both	Estimate + hypothesize + power + sample
Guard	Overgeneralize + p-only + audit + honesty

⚠ FATAL TRAP

Inferential-on-census + 'p = truth' + no-CI/effect + overgeneralize + no-audit = wrong + retraction + shame! Branch + CI + honesty + audit.

★ NORCET HIT

HIT: Parameter + statistic + estimator + unbiased + sufficient + robust + audit!

★ PNC PEARL: Describe all, infer carefully.

► Census + p-values + generalize + junior + bold. Fix trio?

Describe + CI + honesty + branch + mentor.

Nominal + ordinal + interval + ratio + discrete + continuous + dummy + composite + test-match + stats-fit + audit! NOIR + order + distance + zero + operations + tests — level licenses statistics!

Nominal/ordinal/ interval/ratio + audit	Order + distance + zero + operations	Level licenses tests!
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Σ LEVEL — ALLOWS

Level	Allows
Nominal/ordinal	Names/ranks + mode/median + chi + audit
Interval/ratio	Distance/zero + mean/SD + t/ANOVA
Discrete/continuous	Counts/measures + graphs + tests + audit
Guard	Mean-of-nominal + wrong-test + audit + honesty
⚠ FATAL TRAP Mean-of-gender + t-on-ranks + 'numbers are numbers' + wrong-test + no-audit = nonsense + reject + shame! Level + test + audit.	★ NORCET HIT HIT: Stevens + Likert-debate + VAS + semantic + Guttman + audit!

★ PNC PEARL: Level first, test second.

► **Mean-gender + t-ranks + junior + confused + stuck. Fix trio?**

Level + test + mentor + re-analyze.

Tally + classes + width + limits + boundaries + midpoints + exclusive/inclusive + cumulative + graphs + audit! Raw + grouped + ungrouped + range + Sturges + table + chart + interpret + audit!

Tally + classes + limits + midpoints	Exclusive/inclusive + cumulative + graphs	Range + Sturges + audit!
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★ PART — RULES

Part	Rules
Classes/width/limits	Range + k + exclusive + audit + date
Tally/cumulative	Count + cf + interpret + audit + pride
Graphs	Bar/histo/pie + title + source + audit
Guard	Gaps/overlaps + cherry + audit + honesty
⚠ FATAL TRAP Overlapping-classes + 'eyeball' + no-title/source + cherry + no-audit = mess + mislead + reject! Rules + source + audit.	★ NORCET HIT HIT: Sturges + Scott + Freedman + class-mark + ogive + audit!

★ PNC PEARL: Classes clean, counts honest, charts titled.

► **Overlap + no-title/source + cherry + junior + mess. Fix trio?**

Rules + source + honesty + redo + mentor.

GRAPHS: BAR, HISTOGRAM, PIE + MORE

Bar + histogram + polygon + curve + line + pie + ogive + pictograph + map + title + source + scale + audit! Discrete-bars + continuous-histo + parts-pie + time-line + cumulative-ogive!

Bars-discrete +
histo-continuous

Pie-parts + line-
time + ogive-cum

Title + source +
scale + audit!

▲ GRAPH — USES

Graph	Uses
Bar/pie/pictograph	Discrete + parts + pictures + audit
Histogram/polygon/curve	Continuous + shape + normal + audit
Line/ogive/map	Time + cumulative + place + audit
Guard	3D + truncate + cherry + audit + honesty
⚠ FATAL TRAP Pie-everything + 3D + truncated + 'pretty' + no-title/source + cherry = mislead + reject + shame! Right + titled + sourced + audit.	★ NORCET HIT HIT: Tufte + lie-factor + chartjunk + data-ink + excellence + audit!

★ PNC PEARL: Right graph, honest scale, full title.

► 3D-pie + truncated + no-title + junior + flashy. Fix trio?

Right + scale + title + source + redo.

MEAN, MEDIAN, MODE

Mean (sum/n + sensitive!) + median (middle + robust!) + mode (frequent + nominal!) + grouped + weighted + GM/HM + outliers + skew + choose + report + audit! Center + shape + choice + honesty!

Mean-sensitive +
median-robust + mode

Grouped/weighted/
GM/HM + outliers

Skew + choice
+ audit!

◆ AVERAGE — USES

Average	Uses
Mean/SD	Interval + symmetric + infer + audit
Median/IQR	Skewed + ordinal + robust + audit
Mode	Nominal + frequent + quick + audit
Grouped/weighted	Classes + weights + GM/HM + audit

⚠ FATAL TRAP

Mean-everything + 'average' + outlier-blind + skewed-mean + no-median/mode + no-audit = lie + wrong + reject! Shape + choice + audit.

★ NORCET HIT

HIT: Skew + bimodal + uniform + trim + winsorize + audit!

★ PNC PEARL: Shape picks average, honesty reports it.

► Skewed + mean-only + outlier + junior + wrong. Fix trio?

Shape + median + report + honesty + redo.

MEAN VS MEDIAN VS MODE

Normal-mean + skew-median + nominal-mode + income-median + age-mean + categories-mode + report-all + choose + justify + audit! Compare + contrast + defend + document — trio + tale + truth!

Normal-mean + skew-median + nominal	Income-median + age-mean + audit	Compare + justify + document!
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Σ DATA — PICKS

Data	Picks
Normal/symmetric	Mean + SD + t + audit + date
Skewed/ordinal	Median + IQR + ranks + audit
Nominal/categories	Mode + chi + bars + audit
Report	All-three + shape + justify + audit

⚠ FATAL TRAP

One-average + 'average salary' + mean-skewed + no-median + policy-wrong = lie + injustice + shame!
Shape + trio + justify + audit.

★ NORCET HIT

HIT: Mean + housing + CEO-pay + classroom + example + media + audit!

★ PNC PEARL: Report trio, justify choice.

► CEO-pay + mean + policy + skewed + press + angry. Fix trio?

Median + trio + justify + re-report + sorry.

Max-min + IQR + variance + mean-deviation + spread + outliers + SD + audit! Range-quick + variance-average-squared + honest + report + audit — spread tells stability + quality!

Max-min + IQR + variance + audit	Spread + outliers + stability + audit	Report + honesty + audit!
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★ SPREAD — USES

Spread	Uses
Range/IQR	Quick + robust + box + audit + date
Variance/SD	Average-squared + infer + audit
Mean-deviation	Absolute + simple + audit + pride
Guard	Units + outliers + cherry + audit + honesty
⚠ FATAL TRAP Range-only + 'spread fine' + outlier-blind + no-SD/IQR + no-audit = lie + wrong + reject! Spread + shape + audit.	★ NORCET HIT HIT: Sum-squares + df + MS + RMS + CV + audit!

★ PNC PEARL: Range winks, variance verifies.

► Range-only + outlier-blind + junior + claim + spread. Fix trio?

IQR + SD + honesty + report + mentor.

Root-mean-squared + 68-95-99.7 + z-scores + CV + control-limits + bell + audit! Average-distance + comparability + quality + inference — SD speaks deviation fluently!

Root-mean-squared + 68-95-99.7	z + CV + limits + bell + audit	Compare + infer + audit!
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▲ USE — DOES

Use	Does
Describe/compare	Groups + labs + vitals + audit + date
Infer/control	CI + tests + limits + quality + audit
Teach/report	68-95 + z + CV + graphs + audit
Guard	n-small + skew + misuse + audit + honesty
⚠ FATAL TRAP ±-everything + 'normal' + skewed-SD + no-CI/n + misuse + no-audit = wrong + reject + shame! Shape + n + CI + audit.	★ NORCET HIT HIT: Empirical + Chebyshev + SEM + SE + margin + audit!

★ PNC PEARL: SD with shape, CI with n.

► SD-skewed + no-CI/n + junior + claim + wrong. Fix trio?

Shape + n + CI + honesty + redo.

Bell + symmetric + mean-median-mode + 68-95-99.7 + z + tables + tests + QC + audit! Heights + BP + Hb + IQ + errors + nature — normal + common + testable + beautiful!

Bell + symmetric + 68-95-99.7	z + tables + tests + QC	Nature + audit + pride!
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◆ TRAIT — USES

Trait	Uses
Bell/shape	Heights + BP + Hb + IQ + audit + date
z/tables/tests	Scores + areas + t/ANOVA + audit
QC/limits	Labs + process + Westgard + audit
Guard	Skew + small + transform + audit + honesty
⚠ FATAL TRAP Normal-assumed + 'bell always' + skewed-tested + no-transform/check + no-audit = wrong + reject + shame! Test + transform + audit.	★ NORCET HIT HIT: Shapiro + KS + QQ + log + Box-Cox + CLT + audit!

★ PNC PEARL: Test normality, then trust tests.

► **Skewed + t-tested + no-check + junior + publish. Fix trio?**

Check + transform + nonparam + honesty.

TESTS OF SIGNIFICANCE (CONCEPT)

H0 + H1 + alpha + p + power + effect + CI + decide + report + replicate + audit! p + evidence-against-H0 + threshold + context + honesty — significance + sense + science!

H0/H1 + alpha + p + power + effect	Decide + report + replicate + audit	Sense + science + honesty!
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Σ ITEM — RULES

Item	Rules
H0/H1/alpha/p	State + threshold + evidence + decide
Power/effect/CI	Design + size + magnitude + report
Decide/report	Retain/reject + exact + honest + audit
Guard	p-hack + HARKing + spin + audit + registry
⚠ FATAL TRAP $p < 0.05$ + 'proven' + p-hack + HARKing + no-effect/CI + spin + no-register = fraud + retraction + shame! Pre + power + honest + audit.	★ NORCET HIT HIT: ASA + Greenland + Amrhein + abandon + compatibility + s-values + audit!

★ PNC PEARL: p informs, effect matters, honesty rules.

► p-hack + HARKing + spin + junior + publish + pressured. Fix trio?

Pre-register + power + honest + mentor.

Alpha-false-alarm + beta-miss + power + trade-off + sample + effect + consequences + ethics + audit! Reject-true + retain-false + balance + cost + lives + money + honesty!

Alpha-alarm + beta-miss + power	Trade-off + sample + effect + cost	Balance + ethics + audit!
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★ ERROR — MEANS — COSTS

Error	Means	Costs
Type-I/alpha	Reject-true + alarm	Treat + scare + money
Type-II/beta	Retain-false + miss + power	Death + miss + regret
Balance	Sample + effect + ethics + audit	
Report	Power + limits + honesty + audit + date	

⚠ FATAL TRAP

Alpha-only + 'significant' + power-ignored + underpowered + overclaim + no-audit = false + waste + shame! Power + balance + audit.

★ NORCET HIT

HIT: 2×2 + sensitivity + specificity + PPV + ROC + screening + audit!

★ PNC PEARL: Alarm vs miss — price both before trial.

► Underpowered + overclaim + no-power + junior + publish. Fix trio?

Power + balance + honesty + redo + mentor.

Normal + interval + n + variance + t/ANOVA/Pearson + power + assumptions + check + transform + audit! Ranks + free + small + skew + ordinal + sign + median + audit — assumptions audited, alternatives ready!

Normal/interval/n/ t/ANOVA/Pearson	Ranks/free/small/ skew/sign + audit	Check + transform + alternative!
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▲ FAMILY — HOUSES

Family	Houses
Parametric	t + ANOVA + Pearson + regression + audit
Nonparametric	Sign + Wilcoxon + Mann + Kruskal + Spearman
Check/transform	Normal + variance + log + audit + date
Choose	Level + n + shape + consult + audit + honesty

⚠ FATAL TRAP

t-everything + 'robust' + no-check + skewed/small + wrong + no-audit = wrong + reject + shame! Check + alternative + audit.

★ NORCET HIT

HIT: Levene + Bartlett + Fligner + audit!

★ PNC PEARL: Assumptions checked, alternatives known.

► t-skewed + no-check + junior + publish + small. Fix trio?

Check + ranks + honesty + redo + mentor.

t-TEST + CHI-SQUARE + ANOVA (AWARENESS)

Means + categories + groups + t + chi + F + p + assumptions + post-hoc + effect + report + audit!
Paired/unpaired + goodness/independence + one-way/repeated + Bonferroni + eta + audit —
awareness + consult + software!

t-means + chi-cats
+ ANOVA-groups

Assumptions +
post-hoc + effect

Awareness + consult
+ audit!

◆ TEST — USES

Test	Uses
t-test	Means + paired/unpaired + CI + audit
Chi-square	Categories + goodness/independence + audit
ANOVA	Groups + F + post-hoc + effect + audit
All	Assumptions + consult + software + audit

⚠ FATAL TRAP

t-3-groups + chi-small + ANOVA-no-post-hoc + 'p' +
no-assumptions/consult + overclaim = wrong + reject +
shame! Right + check + consult + audit.

★ NORCET HIT

HIT: Welch + Fisher + Yates + McNemar + Cochran +
Friedman + Kruskal + audit!

★ PNC PEARL: Right test, checked assumptions, consulted expert.

► t-3-groups + no-check + junior + publish + bold. Fix trio?

ANOVA + check + consult + honesty + redo.

Relate + strength + direction + scatter + r + cause-NOT + confound + longitudinal + audit! +1/-1/0 + weak/moderate/strong + significant + clinically + honestly — relate + predict, never prove!

+1/-1/0 + weak/moderate/strong	Scatter + r + confound + audit	Predict, never prove cause!
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Σ r — MEANS — ACT

r	Means	Act
+1/-1/0	Perfect/ none + direction + audit + date	
Weak/moderate/strong	0.1/0.3/0.5 + context + CI + audit	
Significant/clinical	p + effect + use + honesty + audit	
Guard	Cause + third + range + outliers + audit	

⚠ FATAL TRAP

r + 'causes' + restricted-range + outlier-driven + no-confounder + overclaim = myth + policy-wrong + shame! Scatter + adjust + honest + audit.

★ NORCET HIT

HIT: Ecological + Berkson + restriction + attenuation + audit!

★ PNC PEARL: Scatter first, r second, cause never (alone).

► r-high + 'causes' + cross + junior + press + bold. Fix trio?

Scatter + confound + design + honesty.

PEARSON + SPEARMAN CORRELATION

Pearson (interval + linear + normal + r + scatter!) + Spearman (ranks + monotonic + ordinal + skew + outliers!) + Kendall + point-biserial + phi + partial + audit! Linear + rank + robust + report!

**Pearson: linear +
interval + normal**

**Spearman: ranks +
monotonic + robust**

**Partial + audit
+ report!**

★ r — USES

r	Uses
Pearson	Linear + interval + normal + scatter + audit
Spearman/Kendall	Ranks + monotonic + skew + outliers
Point-biserial/phi	Dichotomous + categories + audit + date
Partial/multiple	Adjust + confound + regression + audit

⚠ FATAL TRAP

Pearson-skewed + 'strong' + outlier-driven +
no-scatter/ranks + overclaim = wrong + reject + shame!
Scatter + ranks + honest + audit.

★ NORCET HIT

HIT: Fisher-z + CI + power + n + linearity +
homoscedasticity + audit!

★ PNC PEARL: Scatter decides: Pearson or Spearman.

► Skewed + Pearson + outlier + junior + claim + strong. Fix trio?

Scatter + Spearman + honesty + redo + mentor.

PACKAGES + SIGNIFICANCE VS CLINICAL

SPSS + R + Epi-Info + Excel + Stata + SAS + code + output + verify + audit! Statistical (p + math!) vs clinical (effect + cost + patient!) + MCID + NNT + values + shared-decision + audit!

SPSS/R/Epi/Excel +
code + verify

p-math vs effect-
cost-patient + MCID

Values + shared +
audit!

▲ TOOL / GAP — RULES

Tool / gap	Rules
Packages/code	SPSS/R + script + verify + audit + date
p-vs-clinical	Math vs effect + MCID + NNT + values
Decide/share	Patient + cost + ethics + audit + pride
Guard	Black-box + click + no-verify + audit + honesty
⚠ FATAL TRAP Click + 'software says' + no-verify/code + p-only + clinically-trivial + overclaim = wrong + waste + shame! Verify + effect + honesty + audit.	★ NORCET HIT HIT: Reproducible + RMarkdown + Git + version + open + audit!

★ PNC PEARL: Code verified, effect valued, patient centered.

► Click + p-only + trivial + junior + claim + publish. Fix trio?

Verify + effect + honesty + redo + mentor.

COMMUNICATING FINDINGS + REPORT STRUCTURE

Abstract + intro + methods + results + discussion + conclusion + refs + tables + audit! IMRAD + clarity + honesty + limits + implications + recommendations + appendices + audit — write to be used, not shelved!

IMRAD + tables +
refs + audit

Clarity + honesty +
limits + implications

Used, not
shelved!

★ PART — HOLDS

Part	Holds
Abstract/intro/methods	Summary + gap + design + sample + tools
Results/discussion	Tables + compare + limits + honesty
Conclusion/refs	Implications + recommend + cite + audit
Guard	Spin + hide + plagiarize + audit + honesty

⚠ FATAL TRAP

'Thesis done' + spin + hidden-limits + no-implications + shelf + no-publish/use = waste + degree + dust! Write + publish + use + audit.

★ NORCET HIT

HIT: CONSORT + STROBE + PRISMA + COREQ + SQUIRE + CARE + audit!

★ PNC PEARL: Write for users, not shelves.

► Thesis + shelf + no-publish/use + junior + done. Fix trio?

Publish + present + use + mentor + plan.

Title + abstract + keywords + IMRAD + refs + cover + journal + review + revise + publish + audit!
Proposal (problem + review + design + budget + ethics + timeline!) + committee + funds + registry + audit!

Title + IMRAD + refs + journal	Proposal + budget + ethics + timeline	Review + revise + publish + audit!
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▲ PIECE — HOLDS

Piece	Holds
Article/paper	Title + IMRAD + refs + journal + revise
Proposal/thesis	Problem + design + budget + ethics + audit
Both	Clear + honest + cited + formatted + audit
Guard	Salami + duplicate + gift + audit + honesty

⚠ FATAL TRAP

Salami + duplicate + gift + 'publish-or-perish' + predatory + no-ethics + no-audit = retraction + blacklist + shame! Original + journal + audit.

★ NORCET HIT

HIT: ICMJE + COPE + authorship + contributorship + acknowledgments + conflicts + audit!

★ PNC PEARL: One study, one honest paper (usually).

► Salami + duplicate + predatory + junior + pressured. Fix trio?

Original + journal + ethics + mentor + slow.

Read + appraise + adapt + pilot + implement + evaluate + sustain + spread + audit! CASP + JBI + validity + relevance + feasibility + barriers + champions + policy + bedside + audit — appraise + apply + audit!

**Read + appraise +
adapt + pilot**

**Implement + evaluate
+ sustain + spread**

**Champions + policy
+ audit!**

◆ STEP — DOES

Step	Does
Read/appraise	CASP + validity + relevance + audit + date
Adapt/pilot	Context + trial + feedback + revise + audit
Implement/evaluate	Roll + measure + sustain + spread + audit
Guard	Top-down + no-fit + no-evaluate + audit + honesty

⚠ FATAL TRAP

'Evidence-based' + no-appraisal/adapt + copy-paste + no-evaluate/sustain + no-audit = ritual + harm + waste!
Appraise + adapt + evaluate + audit.

★ NORCET HIT

HIT: PARIHS + KTA + RE-AIM + CFIR + Normalization
+ champions + audit!

★ PNC PEARL: Appraised + adapted + audited = adopted.

► **Copy-paste + no-appraisal/adapt + fail + ward + junior. Fix trio?**

Appraise + adapt + evaluate + mentor + pilot.

Team + roles + Gantt + budget + ethics + collect + analyze + report + publish + audit! Authorship + ghost + gift + salami + duplicate + plagiarism + fabrication + COPE + ICMJE + UGC + audit!

Team + Gantt + budget + ethics	Authorship + COPE + ICMJE + UGC + audit	No-ghost/gift/ salami + audit!
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Σ ITEM — RULES

Item	Rules
Project/team	Roles + Gantt + budget + ethics + audit
Authorship/ethics	COPE + ICMJE + gift/ghost + audit + date
Guard	Salami/duplicate/plagiarism + audit + honesty
All	Report + publish + use + audit + pride
⚠ FATAL TRAP Gift/ghost + 'professor first' + no-criteria/consent + junior-exploited + no-audit = fraud + resentment + retraction! Criteria + consent + audit.	★ NORCET HIT HIT: CRediT + ORCID + contributorship + order + corresponding + audit!

★ PNC PEARL: Credit earned, never gifted or ghosted.

► **Gift-first + junior-last + no-criteria + tears + lab. Fix trio?**

Criteria + consent + CRediT + mentor + redo.

PREDATORY JOURNALS + DATA MANAGEMENT

Flattery + fees + fast + fake + no-review + blacklist + Beall + DOAJ + IF + ISSN + audit! Plan + collect + clean + store + backup + share + retain + destroy + audit — publish reputable, preserve responsibly!

Flattery/fees/fast/
fake = predatory

DOAJ + IF +
ISSN + audit

Plan + backup +
share + audit!

★ CHECK — DOES

Check	Does
Journal	DOAJ + IF + board + fees + peers + audit
Data-plan	Collect + clean + store + backup + share
Retain/destroy	Period + consent + privacy + audit + date
Guard	Predatory + loss + breach + audit + honesty

⚠ FATAL TRAP

Pay-publish + 'international' + no-DOAJ/board + thesis-lost + data-leak + no-audit = scam + loss + breach + shame! Verify + backup + audit.

★ NORCET HIT

HIT: UGC-CARE + Scopus + WoS + PubMed + Q + H-index + DORA + Leiden + audit!

★ PNC PEARL: Reputable journal, retrievable data.

► Pay-publish + flattery + no-DOAJ + junior + eager. Check trio?

DOAJ + board + peers + mentor + slow.